STAT 4830 - SPRING 2026

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Success and Failure Analysis of Evolutionary Strategies (ES) in LLM fine-tuning

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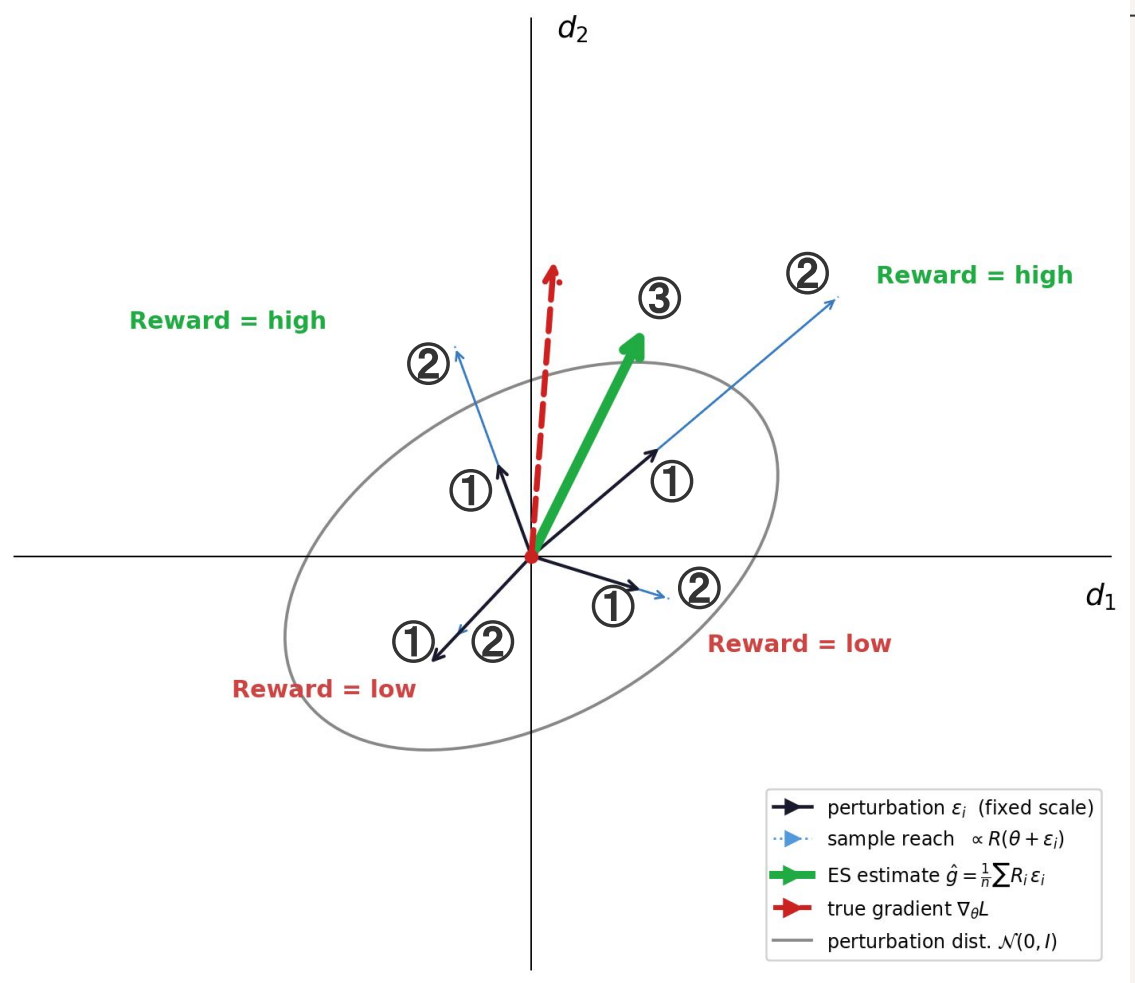
What is ES?

- ① Throw random variations of your policy into the world, see which ones do better
- ② Give more credit to the directions that scored high rewards and less to the ones that flopped
- ③ Average those weighted directions → you get a "sense" of which way to improve
- Repeat. That's it — no gradients, no backprop, just reward signals



It’s the reward function, stupid!

Cross Entropy Reward Function

$$r(x, y^*) = \log P_\theta(y^* \mid x)$$

- Continuous
- Dense gradient signal (magnitude represents confidence)
- Low variance and high sample efficiency
- **Requires white box logit access**

Binary Reward Function

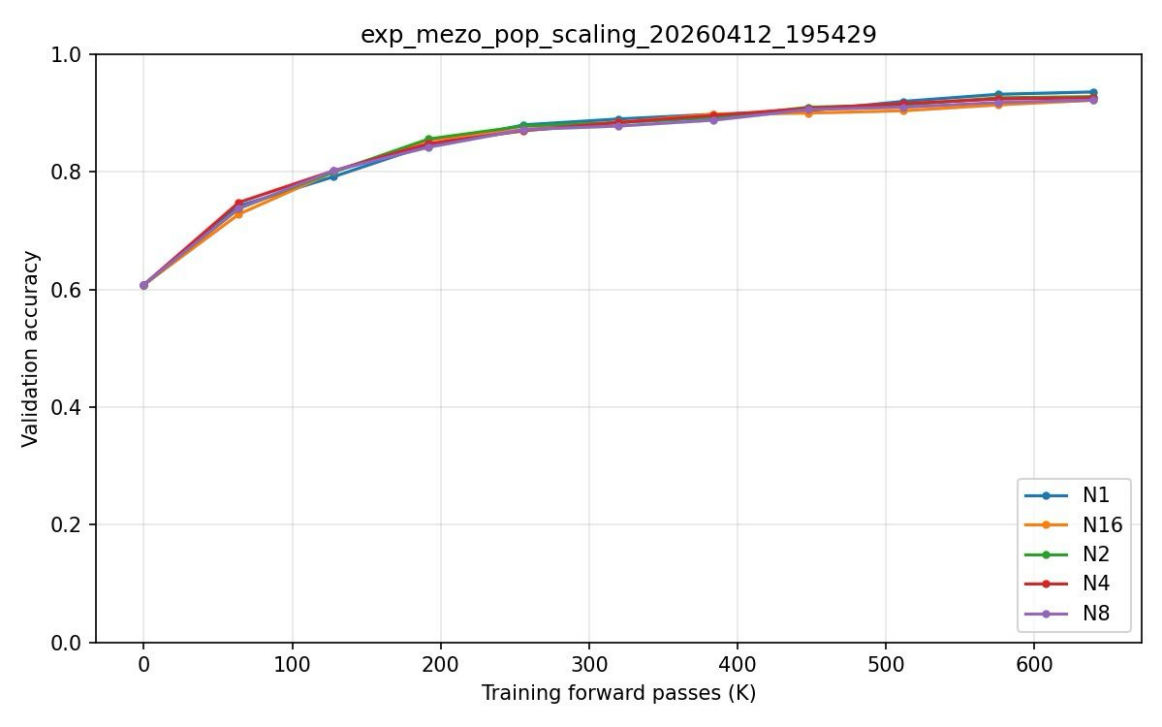
$$r(y, y^*) = \begin{cases} 1 & \text{if } y = y^* \\ 0 & \text{otherwise} \end{cases}$$

- Discrete
- Sparse gradient signal (only direction, no magnitude)
- High variance and low sample efficiency
- **Generation only - black box**

It’s the reward function, stupid!

Cross-entropy Loss

- Works even when N=1 (and N-agnostic, if learning rate is scaled accodingly)
- Exact estimate of gradient
- Very favorable performance
- Limitation: Black-box assumption is broken since gaining logit assumes white-box access

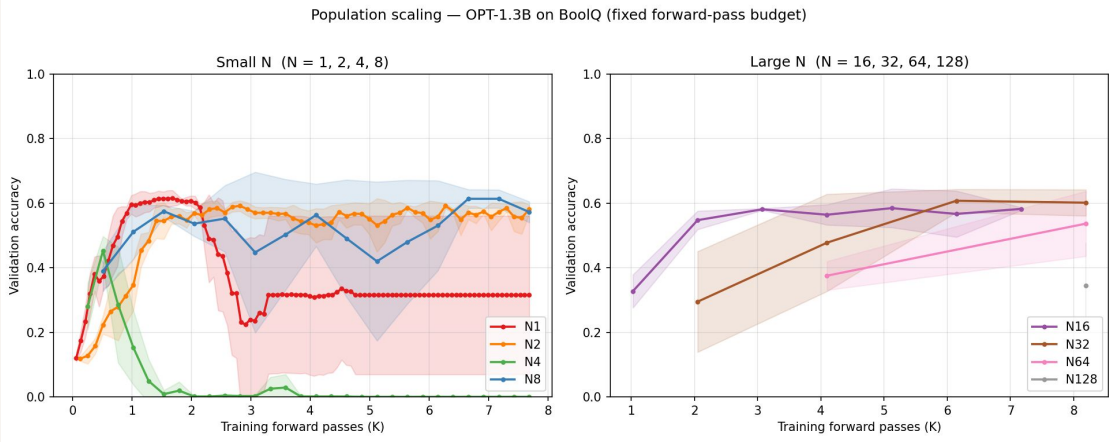


OPT 13B Fine Tuning on SST2 (binary classification task)
Notice the population indifference to accuracy convergence

It’s the reward function, stupid!

Binary Loss

- Not N agnostic - we observe that the model is likely to degrade at low N
- This is likely due to how binary rewards don’t provide a clear signal on how the weights should be perturbed
- Binary reward functions are most applicable to fine tuning situations



OPT 1.3B Fine Tuning on BoolQ (binary classification task)
Notice the population difference to accuracy convergence

Two Faces of ES - benign and malignant regime

Benign Regime (format errors dominate)
- Model reasons correctly but outputs wrong structure (e.g. missing #### <number>)
- Format errors are systematic — same mistake across examples → consistent reward signal
- ES detects the fix reliably even at N=2
- No normalization acts as a signal filter: near-zero advantages → near-zero updates → pretrained reasoning preserved
- Result: stable, monotonic improvement

Malignant Regime (answer errors dominate)
- Model reaches the wrong conclusion regardless of format
- Answer errors are stochastic — different reasoning failure per example → conflicting, near-zero advantages
- No perturbation fixes example A without hurting example B
- ES gradient estimate points nowhere useful — larger N doesn't help
- Result: ES stalls or degrades

Two Faces of ES - benign and malignant regime

Why Normalization Flips Benign → Malignant

- Normalization amplifies near-zero advantages to full-magnitude updates
- In benign regime: turns a stable filter into a destructive random walk
- High-dimensional geometry (~1.5B params): nearly every perturbation is orthogonal to the true gradient — normalization amplifies this orthogonal noise

Core Claim

> ES is a format-correction machine, not a reasoning engine. Small N + no normalization is the right configuration precisely because it matches the sparse, systematic signal
> structure of the benign regime.

Limitations & Next Steps

- Results limited to Qwen 2.5 1.5B Instruct on a narrow task set — generalizability across models and tasks remains unverified
- Open question: Can ES improve reasoning capabilities, or is it fundamentally limited to format correction?
    - Characterize the malignant regime: how does ES degrade when answer errors dominate — regression, collapse, or random walk?
    - Identify the configuration boundary: minimum N, normalization, and lr where benign-regime gains hold vs. break down
    - Test the benign/malignant decomposition explicitly by controlling the ratio of format vs. answer errors at training time

THANK YOU

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